

Auteur: Dara van Engelen
Studierichting: Informatica
Oefeningenreeks 1B nr 43.10

$$9z^2 + 3(2i - 5)z + 9 - 7i = 0$$

$$\Leftrightarrow 9z^2 + (6i - 15)z + 9 - 7i = 0$$

$$D = (6 - 5i)^2 - 4 \cdot 9 \cdot (9 - 7i) = 36i^2 - 180i + 255 - 324 + 252i$$

$$= -36 - 180i + 225 - 324 + 252i = -135 = 72i$$

$$-135 + 72i = (x + yi)^2 = x^2 - y^2 + 2xyi$$

$$\begin{cases} x^2 - y^2 &= -135 \\ 2xy &= 72 \end{cases}$$

$$\Leftrightarrow \begin{cases} x^2 - y^2 &= -135 \\ x^2(-y)^2 &= -1296 \end{cases}$$

$$\Rightarrow \lambda^2 + 135\lambda - 1296 = 0$$

$$D = 135^2 - 4 \cdot 1 \cdot (-1296) = 18225 + 5184 = 23409$$

$$\lambda_1 = \frac{-135 + 153}{2} = 9 = x^2 \Rightarrow x = \pm 3$$

$$\lambda_2 = \frac{-135 - 153}{2} = -144 = -y^2 \Rightarrow y = \pm 12$$

$$\Rightarrow (x + yi) \in \{3 + 12i, -3 - 12i\}$$

$$z_1 = \frac{-6i + 15 + 3 + 12i}{18} = \frac{6i + 18}{18} = \frac{i}{3} + 1$$

$$z - 2 = \frac{-6i + 15 - 3 - 12i}{18} = \frac{-18i + 12}{18} = -i + \frac{2}{3}$$

References